ARCHIGEN-3D / MODEL EVIDENCE REPORT

Floorplan and facade evaluation

Benchmark matrices, GitHub experiment history and MLflow results
10 November 2026 · GitHub main a30ef02

Floorplan segmentation and reconstruction

The registered ResNet34 UNet++ champion reaches **74.17% merged-wall IoU**, **61.78% door IoU** and **56.40% 10-class room mIoU** on the 400-plan CubiCasa5K test split. Four-fold test-time augmentation raises all three to **74.93%**, **63.21%** and **58.24%**, with measured inference rising from **104 to 328 ms** per plan. On the locked 12-plan external set, the champion produced **9 valid shells** and averaged **5.25 room-count errors** per plan.

Facade synthesis

On the historical CMP test protocol, the registered 256-pixel Pix2Pix EMA checkpoint has the lowest recorded perceptual distances, with **LPIPS 0.3096** and **KID ×1000 of 63.552**. The experimental 512-pixel model improves LPIPS from **0.49887 to 0.49241** in a separate paired test against the same EMA model. The experimental results cover unlabelled CMP facades.

Coverage

Source	Coverage
GitHub	17 published heads, including 9 experiment heads. Four additional local experiment refs are identified separately.
Local MLflow	24 runs across 4 populated experiments. Includes all 247 latest metrics and 4,980 historical metric points.
Saved evaluations	In-domain test and validation, external floorplans, facade checkpoints, 512-pixel ablations and diffusion pilots.
Embedded evidence	Open the PDF attachments pane for the ZIP with CSV histories, JSON benchmarks and a source manifest.

This is an audit of recorded measurements. Branch heads, training runs, evaluation imports and deployable checkpoints are linked where the evidence supports a match.

Repository · ayebkalil/archigen-3d · Audited main commit

Model architectures and experiment changes
See pages 19–25 for model diagrams and changes tested on each branch.

Evaluation charts
See pages 26–29 for test comparisons, generalization and training curves.

01 / FLOORPLAN

CubiCasa5K official test

400 plans · ResNet34 UNet++ champion · paired single-pass and four-fold TTA

Segmentation and reconstruction

Metric	Single pass	Four-fold TTA	Change [95% CI]	p
Merged Wall IoU ↑	74.17	74.93	+0.76 [0.63, 0.91]	<0.001*
Interior Wall IoU ↑	62.18	63.25	+1.07 [0.87, 1.28]	<0.001*
Exterior Wall IoU ↑	73.86	74.58	+0.73 [0.52, 0.94]	<0.001*
Door IoU ↑	61.78	63.21	+1.43 [1.11, 1.79]	<0.001*
Window IoU ↑	73.52	74.65	+1.14 [0.77, 1.59]	<0.001*
Struct 3-Class mIoU ↑	69.82	70.93	+1.11 [0.90, 1.34]	<0.001*
Struct 4-Class mIoU ↑	67.83	68.92	+1.09 [0.90, 1.30]	<0.001*
Room Pixel Acc ↑	87.50	88.09	+0.59 [0.10, 1.05]	0.008
Room mIoU (9c, ex. Office) ↑	62.66	64.72	+2.05 [1.29, 2.75]	<0.001*
Room mIoU (10c, inc. Office) ↑	56.40	58.24	+1.85 [1.16, 2.48]	<0.001*
Pred Perimeter Rel Error ↓	24.37	24.20	-0.18 [-2.54, 2.40]	0.462
Pred Perimeter MAE (m) ↓	13.46	13.04	-0.42 [-1.81, 1.03]	0.289
Pred Footprint Rel Error ↓	54.67	55.41	+0.74 [-1.20, 2.90]	0.792
Pred Footprint MAE (m2) ↓	21.69	21.89	+0.19 [-0.56, 1.01]	0.682
Shell Validity Rate ↑	90.75	91.75	+1.00 [-2.75, 4.50]	0.339
Room Instance Prec ↑	4.46	4.31	-0.15 [-0.48, 0.17]	0.816
Room Instance Rec ↑	5.02	4.84	-0.18 [-0.55, 0.19]	0.847
Room Instance F1 ↑	4.73	4.56	-0.17 [-0.52, 0.18]	0.819
Room Count MAE ↓	1.56	1.53	-0.02 [-0.12, 0.07]	0.343
Door Count MAE ↓	1.47	1.44	-0.03 [-0.11, 0.06]	0.294

Inference averaged **104.18 ms** single pass and **327.59 ms** with TTA. The four predictions use the original plan and horizontal, vertical and combined flips. The evaluator inverts the transforms and averages class probabilities.

IoU, rates, precision, recall and F1 are percentages. Their changes are percentage points. The relative errors use percentages, and absolute errors use meters, square meters or counts as indicated. Room-instance matching requires IoU ≥ 0.50 and identical semantic class.

* The evaluator stores p = 0 where none of 1,000 paired bootstrap samples support the opposite direction. The display uses <0.001 to convey sampling resolution. Full single-pass, TTA and difference intervals are in the embedded JSON.

Evidence · official_test_split_authoritative.json · evaluate_test_authoritative.py

02 / FLOORPLAN

Validation and class breakdown

400 validation plans · same champion checkpoint · IoU and accuracy in percent

Validation metrics

Metric	Single pass	Four-fold TTA	Change [95% CI]	p
Merged Wall IoU ↑	73.90	74.32	+0.42 [0.29, 0.56]	<0.001*
Interior Wall IoU ↑	62.25	62.82	+0.57 [0.32, 0.78]	<0.001*
Exterior Wall IoU ↑	73.58	73.99	+0.41 [0.20, 0.62]	<0.001*
Door IoU ↑	61.13	62.02	+0.90 [0.62, 1.20]	<0.001*
Window IoU ↑	74.72	75.48	+0.77 [0.52, 1.05]	<0.001*
Struct 3-Class mIoU ↑	69.91	70.61	+0.70 [0.55, 0.87]	<0.001*
Struct 4-Class mIoU ↑	67.92	68.58	+0.66 [0.51, 0.81]	<0.001*
Room Pixel Acc ↑	89.66	90.61	+0.95 [0.45, 1.46]	<0.001*
Room mIoU (10c) ↑	62.39	63.95	+1.56 [0.91, 2.13]	<0.001*
Room Instance Prec ↑	6.03	6.48	+0.45 [-0.11, 1.07]	0.065
Room Instance Rec ↑	6.80	7.21	+0.41 [-0.23, 1.09]	0.120
Room Instance F1 ↑	6.39	6.83	+0.44 [-0.17, 1.08]	0.089
Room Count MAE ↓	1.50	1.41	-0.09 [-0.17, 0.00]	0.033
Door Count MAE ↓	1.38	1.34	-0.05 [-0.12, 0.03]	0.133
Shell Validity Rate ↑	99.75	99.75	+0.00 [0.00, 0.00]	1.000

Mean validation inference was 61.72 ms single pass and 185.25 ms with TTA.

Room IoU by class on both splits

Class	Test single	Test TTA	Val single	Val TTA
LivingRoom	74.36	75.64	78.37	80.09
Bedroom	74.76	74.71	78.81	80.80
Kitchen	67.52	67.94	73.78	75.38
Bath	69.80	71.75	72.84	74.36
Entry	61.44	63.07	67.20	69.74
Outdoor	72.95	73.92	71.77	73.89
Storage	42.65	45.28	47.61	50.74
Closet	62.43	66.85	71.50	73.73
Garage	38.07	43.28	62.05	60.80
Office	0.00	0.00	0.00	0.00

The nine-class test mean excludes Office. Office IoU is 0 on both splits in this benchmark. The embedded files provide the paired statistics for every class.

Evidence · official_test_split_authoritative.json · val_400_tta_benchmark.json (local evidence)

03 / FLOORPLAN

Champion, challenger and router

Separate 400-plan test and validation comparisons with their own shell-validity checks

Official test · 400 plans

Metric	Champion v1.1	Blend α=0.3	Challenger	Routed system
Room mIoU, 10 classes (%)	56.40	55.41	52.30	56.35
Room mIoU, 9 classes (%)	62.66	61.57	58.12	62.61
Room mIoU, 11 classes (%)	51.27	50.37	47.55	51.23
Merged wall IoU (%)	74.17	74.52	74.27	74.17
Mean wall IoU (%)	68.02	68.27	67.79	68.02
Exterior wall IoU (%)	73.86	74.15	73.79	73.85
Interior wall IoU (%)	62.18	62.38	61.80	62.18
Door IoU (%)	61.78	62.19	61.66	61.77
Window IoU (%)	73.52	73.63	72.34	73.51
Room count MAE	1.562	1.643	1.915	1.575
Room count signed bias	0.022	0.352	0.980	0.040
Valid shells (%)	99.25	99.50	100.00	99.75
Fallback triggers (%)	0.00	0.00	0.00	1.25
Fallback triggers (count)	0	0	0	5

Official validation · 400 plans

Metric	Champion v1.1	Blend α=0.3	Challenger	Routed system
Room mIoU, 10 classes (%)	62.39	61.35	56.94	62.31
Room mIoU, 9 classes (%)	69.33	68.17	63.27	69.23
Room mIoU, 11 classes (%)	56.72	55.77	51.77	56.64
Merged wall IoU (%)	73.90	74.15	73.90	73.90
Mean wall IoU (%)	67.92	68.16	67.68	67.92
Exterior wall IoU (%)	73.58	73.87	73.47	73.59
Interior wall IoU (%)	62.25	62.44	61.89	62.25
Door IoU (%)	61.13	61.49	60.88	61.13
Window IoU (%)	74.72	74.84	73.70	74.72
Room count MAE	1.485	1.583	1.907	1.488
Room count signed bias	-0.030	0.338	0.868	-0.013
Valid shells (%)	99.75	100.00	100.00	100.00
Fallback triggers (%)	0.00	0.00	0.00	1.25
Fallback triggers (count)	0	0	0	5

Routing invoked the α=0.3 fallback on 5 of 400 plans per split. On test it raised shell validity by 0.50 percentage points while room-count MAE rose from 1.562 to 1.575. Shell rates here use the routing benchmark's validity checks, which differ from the TTA benchmark.

Evidence · authoritative_test400_benchmark.json (local evidence) · authoritative_val400_benchmark.json (local evidence)

04 / FLOORPLAN

Geometry and opening diagnostics

CubiCasa5K validation · shell validity, door centroids and perimeter quality

Door detection in pixel space · 400 plans

Matching radius	TP / FP / FN	Precision	Recall	F1	Count MAE
12 px	3,356 / 367 / 432	90.14%	88.60%	89.36%	—
24 px	3,397 / 326 / 391	91.24%	89.68%	90.45%	1.12

Centroid error was 1.77 px mean, 1.12 px median and 2.72 px SD. The radius-based F1 scores measure localized detections. The test door IoU measures pixel overlap.

Footprint and perimeter · 400 plans

Metric	Recorded result
Valid shell	399 / 400 (99.75%)
Footprint IoU	82.16% filled shell · 78.63% raw pixel union
Perimeter absolute relative error at 0.03 m/px	23.94% unpruned · 21.57% pruned
Perimeter median relative error	9.24% unpruned · 9.15% pruned
Perimeter signed relative error	13.69% unpruned · 9.87% pruned
Estimated scale	0.03419 ± 0.01196 m/px (mean ± SD)

Repair protocols

Protocol	Single pass	TTA	Fallback	Repair detail
Topology decomposition	93.25%	94.25%	98.75%	25/25 repairs succeeded
Explicit make_valid comparison	99.25%	99.50%	99.75%	24 single-pass repairs, 20 TTA repairs

The decomposition recorded 79.05% footprint IoU and 50.75% footprint area error. Smoothing reduced mean perimeter error from 23.11% to 21.28%. Estimated physical dimensions depend on a nominal door-width assumption.

Evidence · rigorous_validation_benchmark_results.json (local evidence) · perimeter_decomposition_and_tta_validity.json · validity_with_make_valid_comparison.json (local evidence) · rigorous_validation_benchmark.py

05 / FLOORPLAN

External generalization

External plans, CAD restyling and synthetic stress tests

Locked blind external set · 12 plans

Metric	Champion v1.1	Restyled challenger
Watertight shells	75.0% (9/12)	100.0% (12/12)
Room count MAE	5.25	4.42
Room count signed bias	-2.92	0.58
Door count MAE	8.08	6.92
Door count signed bias	0.08	2.42

The README records 1 of 12 champion outputs usable, 4 needing fixes and 7 failed. The challenger reduced count MAE but increased signed door over-counting. On the official test split its 10-class room mIoU fell from 56.40% to 52.30%.

Development set · 20 plans

Model	Shells	Rooms	Doors	Heuristic usable / fixes / unusable
Champion v1.1	20/20	118	203	11 / 3 / 6
Restyled challenger	20/20	188	203	17 / 1 / 2
Blend α=0.3	19/20	111	218	13 / 1 / 6

Blind ratings artifact	Usable	Needs fixes	Unusable
Champion v1.1	9 (45%)	4	7
Restyled challenger	14 (70%)	4	2

The saved blind ratings favored the challenger 10 times, the champion 4 times, with 6 ties. A separate routed-system run found 110 vs 109 rooms and triggered the fallback 1 time.

Other external checks

Set	Result
Real external archive	12 plans · 75.0% valid shells · 100.0% had detected rooms
Held-out modern plans	8 plans · 6 valid shells single pass and 6 with TTA · 53 rooms in each mode
Synthetic stress set	10/10 failed the saved stress criteria. Failures include missing rooms, under-segmentation and incomplete shells.

Evidence · fresh_defense_evaluation_report.json (local evidence) · fresh_defense_evaluation_report.json (local evidence) · dev_human_ratings.json (local evidence) · real_external_evaluation_report.json · heldout_modern_eval_summary.json (local evidence) · synthetic_stress_benchmark_results.json (local evidence) · README.md

06 / FLOORPLAN

Training and runtime evidence

Alignment fix, CAD fine-tuning and deployment parity

Completed training evaluations in MLflow

Training run	ID prefix	Test merged wall IoU	Test door IoU	Test room mIoU
Early baseline, 15 epochs	50ab5d70	—	10.33%	34.93%
Initial 25-epoch run	39254ff5	12.08%	0.40%	0.78%
Extended baseline, 60 epochs	6bec9eef	22.40%	10.75%	37.90%
Corrected alignment, 25 epochs	112b8d10	74.32%	62.38%	54.73%

Training metrics use each run's logged evaluator. The repo documents a correction to SVG rasterization under 512-pixel letterboxing. The 12-epoch CAD augmentation and six-epoch class-weighted Lovász experiments also appear in the MLflow register. Their logged validation metrics use different metric keys and units.

Restyling and blending on validation

Variant	Room mIoU (%)	Structure mIoU (%)	Room accuracy (%)	Door IoU (%)	Window IoU (%)
Champion v1.1	56.72	67.92	92.31	61.13	74.72
Blend a=0.3	55.77	68.16	91.61	61.49	74.84
Blend a=0.5	54.73	68.17	90.99	61.50	74.76
Blend a=0.7	53.61	68.04	90.38	61.36	74.56
Challenger	51.77	67.48	89.60	60.88	73.70

This comparison excludes background from its room mIoU and includes Other. Class-level IoUs for all five variants are in the embedded file. The restyle training artifact records one epoch with validation room mIoU of 56.94% after starting at 62.39% under its 10-class evaluation.

Runtime and export verification

Check	Recorded result
PyTorch vs ONNX · 50 plans	98.39% mean room overlap · 100% label concordance
Mean count differences	0.00 rooms · 0.02 walls
Mean latency in parity test	344.4 ms PyTorch · 868.9 ms ONNX
Pipeline latency snapshot	175.25 ms single pass · 249.93 ms dual pass

Evidence · per_class_room_iou_comparison.json (local evidence) · restyled_training_metrics.json (local evidence) · onnx_vs_pytorch_comparison.json (local evidence) · pipeline_latency_measurement.json (local evidence) · defense_runbook.md

07 / FACADE

CMP Facades · 256 pixels

Historical held-out comparison · 400 training pairs and 106 test images

Checkpoint	Upsampling	L1 ↓	SSIM ↑	PSNR dB ↑	LPIPS ↓	KID ×1000 ↓
Baseline 100 epochs	transpose	0.3483	0.2866	13.19	0.6327	150.357 ± 10.248
Enhanced 150-epoch checkpoint	nearest	0.3379	0.3345	13.15	0.4940	109.857 ± 9.238
Perceptual v1.0	nearest	0.3526	0.2792	12.69	0.3169	67.649 ± 5.946
Perceptual EMA + schedule	nearest	0.3403	0.2893	12.97	0.3096	63.552 ± 5.826

Interpretation

The enhanced checkpoint leads on L1 and SSIM. The baseline narrowly leads on PSNR. The promoted EMA checkpoint leads on LPIPS and KID. Relative to the baseline, its LPIPS falls 51.1% and its KID falls 57.7% under this protocol.

The standard KID benchmark used seed 42, subsets of 50 and 100 repetitions. The ± values are the reported subset standard deviations. The README separately reports a paired resampling study over 106 images with 2,000 draws, with KID ×1000 at 63.21 ± 5.76 for EMA and 66.90 ± 6.22 for perceptual v1.0. The full per-image record for that historical comparison was unavailable in the audited evidence.

Checkpoint and tracking caveats

The Enhanced branch configured 200 epochs. Its local MLflow run is marked RUNNING and has latest metrics through step 171. The checkpoint compared in this table is the retained 150-epoch checkpoint. The perceptual 100-epoch run reuses the Enhanced run name in MLflow.

The registered facade reference is the EMA checkpoint. README reports a maximum ONNX parity delta of 4.13×10^{-6}. The defense runbook quotes 6.31×10^{-6} without clearly identifying the same artifact and comparison. The two values are reported separately.

Registered model	Checkpoint SHA-256 prefix	Registry role
Floorplan UNet++	80fa1ea7780f7cb2	Champion with blend fallback
Facade Pix2Pix EMA	4d4ec134344564cd	Promoted 256-pixel champion

Evidence · facade_authoritative_metrics.json · model_registry.json · README.md · defense_runbook.md · evaluate_facade_models.py

08 / FACADE

Paired 512-pixel experiment

CMP test · 106 paired images · identical semantic inputs · EMA outputs upsampled to 512

Metric	EMA 256 at 512	Coarse path	Experimental 512	Experiment − EMA
LPIPS ↓	0.49887	0.50589	0.49241	-0.00646
SSIM ↑	0.33057	0.33203	0.36176	+0.03119
PSNR (dB) ↑	13.069	12.758	13.329	+0.259
Opening-edge MAE ↓	0.10771	0.10504	0.09493	-0.01277
Torchvision KID ×1000 ↓	69.944	74.673	69.616	-0.329

The paired 95% bootstrap interval for experimental-minus-EMA LPIPS is **[-0.01014, -0.00264]**. Opening-edge MAE fell 11.86%. The small KID change is descriptive. This test uses 512-pixel targets and a full-set Torchvision Inception estimator. Use its EMA row for comparison within this test.

Controlled training and checkpoint selection

Training variant	Epochs logged	Best val VGG	Best epoch	Peak device MiB
Initial residual head	4	1.12398	1	2342
Higher VGG and edge weights	5	1.08337	1	2129
Coarse decoder fine-tuned	5	1.05881	1	2081

The selected checkpoint is decoder epoch 1, chosen by validation VGG before the test. The residual FiLM model uses two PatchGAN scales, gradient checkpointing, physical batch 2 and four-step accumulation for effective batch 8. CMP supplies no labels for the four requested styles. The API reported neutral synthesis.

Integration checks recorded by the experiment

Check	Recorded result
Evaluation memory	1246 MiB sampled device peak · 737.7 MiB PyTorch allocation peak
ONNX vs PyTorch	3.16×10^{-6} maximum absolute difference, per experiment report
Viewer frame and UV alignment	≤0.01925 px at 512 · ≤0.00496 px in Phase 1
Saved verification	79 Python tests · 13 frontend geometry tests · Vite build passed

The viewer checks cover semantic frame/UV alignment on L, T, U and detached fixtures. Image details require a separate visual assessment. The experiment is recorded as unpromoted and still shows grid artifacts and softness.

Evidence · test_metrics.json · experiment_results.json · verification.json (local evidence) · facade512-experiment.md

09 / FACADE

512 training logs and diffusion

Saved ablations and experimental 1024-pixel material trials

All recorded 512 training epochs

Variant	Epoch	Step	Loss G	Loss D	L1	VGG	Edge	Val VGG	Peak MiB
Initial	1	50	4.0946	0.9609	0.3183	1.1038	0.1041	1.12398	1810
Initial	2	100	4.3386	0.8581	0.3232	1.1041	0.1050	1.14702	1932
Initial	3	150	4.5050	0.8654	0.3381	1.1286	0.1109	1.17013	2342
Initial	4	200	4.5988	0.8948	0.3464	1.1472	0.1141	1.18534	2342
VGG + edges	1	50	13.8480	0.9376	0.3120	1.0637	0.0993	1.08337	1872
VGG + edges	2	100	14.1292	0.7334	0.3158	1.0521	0.0995	1.08696	2129
VGG + edges	3	150	14.4098	0.6888	0.3275	1.0573	0.1064	1.09219	2129
VGG + edges	4	200	14.5630	0.7043	0.3355	1.0632	0.1096	1.09743	2129
VGG + edges	5	250	14.6248	0.7431	0.3405	1.0656	0.1126	1.09747	2129
Decoder	1	50	13.5073	0.9264	0.3106	1.0323	0.0956	1.05881	1994
Decoder	2	100	13.9514	0.7028	0.3135	1.0322	0.0983	1.07598	2081
Decoder	3	150	14.2842	0.6931	0.3250	1.0461	0.1064	1.07779	2081
Decoder	4	200	14.4040	0.7127	0.3314	1.0514	0.1092	1.08838	2081
Decoder	5	250	14.4678	0.7711	0.3366	1.0549	0.1105	1.08415	2081

SDXL + Canny control and material composite

Case	Control scale	Steps	Elapsed s	Sampled peak MiB	Review
Viewer geometry trial	1.0	30	444.8	795	Rejected
Brick material prototype	0.0	30	427.7	824	Material prototype
CMP validation pilot	0.5	30	229.8	623	Rejected
Viewer geometry trial	0.5	30	428.2	839	Rejected
Viewer geometry trial	1.0	30	630.0	800	Rejected

The 1024-pixel composite preserved 5 prescribed openings. The maximum fixture frame/UV error was 0.03849 px. Review noted sharper detail and visible mirrored tiling and image-based glass. No held-out LPIPS, SSIM, PSNR, FID or KID was reported for diffusion.

The sampled device counter and PyTorch counters come from different instruments. PyTorch recorded 997.3 MiB peak allocated and 1,134 MiB reserved for the pilot runs.

Evidence · experiment_results.json · facade-diffusion-results.json · facade-diffusion-experiment.md

10 / PROVENANCE

GitHub experiment register

Published experiment branches, linked to MLflow records and retained evaluations

GitHub branch	Tip	Experiment	MLflow / other evidence
experiment/baseline-pix2pix-100ep	77313d5	Facade · baseline Pix2Pix 100 epochs	e463b47e trained on earlier main. Historical CMP matrix supplies test metrics.
experiment/enhanced-pix2pix-200ep	352a76e	Facade · Enhanced upsampling	8de9e858 training snapshot through step 171. The saved CMP comparison uses the 150-epoch checkpoint.
experiment/perceptual-sharpness-pix2pix	36ec4e3	Facade · perceptual Pix2Pix v1.0	1ea21de0 logged the 100-epoch run. Held-out results are in the historical CMP matrix.
experiment/model-refinement-and-robustness	883ee19	Facade · EMA and schedule. Floorplan · later refinement	702efec3 facade training and cc26ef57 floorplan training. EMA release metrics were imported as 1835cac5.
experiment/floorplan-style-adaptation	acfd336	Floorplan · CAD restyle, interpolation and routing	cfee34c9 is a fallback metadata import. Restyle training, blend ablations and blind results are in saved JSON.
experiment/facade-photorealism-512	3dc23fc	Facade · residual FiLM and dual-scale PatchGAN	ee9e3b8a imports selected test metrics and decoder history. Three training ablations are retained.
experiment/facade-diffusion-control	791d2b3	Facade · SDXL + Canny and material composite	ee444a71 imports 1024-pixel resolution and control steps. Five pilot outcomes are in the branch report.
experiment/furniture-reconstruction-quality	5b32a0a	Furniture reconstruction experiment	Different model family. Outside the two-model evaluation.
experiment/studio-interface-design	17b9686	Frontend design experiment	Interface changes. No separate floorplan or facade run was identified.

Related published refs

Feature branch **feature/floorplan-segmentation-3d** is the source tag for the early floorplan training series, CAD fine-tune and Lovász run. **release/v1.0-defense-models** tags the champion imports a7e87552 and 1835cac5. Early facade smoke runs are tagged main. The evidence ZIP lists every fetched public head.

Additional local experiment refs

Local branch	Tip	Related evidence
experiment/floorplan-baseline-15ep	0b67f07	Early baseline family · 50ab5d70
experiment/floorplan-extended-60ep	186c8fb	Initial 25-epoch attempt and extended 60-epoch family · 39254ff5, 3a3605b6, 6bec9eef
experiment/floorplan-aligned-25ep	b0b8399	Alignment-corrected family · 0dfc70e5, 4ee5f64d, 112b8d10
experiment/house-geometry-reconstruction	7bcf84a	Downstream geometry repair and viewer checks. No separate trained model run identified.

A tip SHA identifies the branch snapshot. An MLflow source commit identifies the code recorded at execution. For historical runs the report also uses training configuration and checkpoint lineage.

Published refs · GitHub branch browser

11 / MLFLOW

Floorplan MLflow · early

floorplan-multitask-unetpp

Latest recorded values in original logged units. Brackets give the logged step. Step 0 is common for one-off evaluation imports. Run IDs and source tags are shown under each heading.

Early one-epoch attempt · FAILED

0e8168367721436d914475797dd4672c · feature/floorplan-segmentation-3d · f59bd2f · 1 epochs configured

No metrics recorded.

Early one-epoch smoke run · FINISHED

832abe91c70943d5874ef048b195eb33 · feature/floorplan-segmentation-3d · f59bd2f · 1 epochs configured

learning_rate 1e-06 [1] · **test_door_iou** 0.008691873 [0] · **test_room_acc** 0.02664746 [0] · **test_room_miou** 0.005444557 [0] · **test_struct_miou** 0.01182599 [0] · **test_val_loss** 5.661918 [0] · **test_wall_ext_iou** 0.0385224 [0] · **test_wall_int_iou** 8.967572e-05 [0] · **test_win_iou** 0 [0] · **train_loss** 5.486902 [1] · **train_room_loss** 3.026016 [1] · **train_struct_loss** 3.066089 [1] · **val_door_iou** 0.01555471 [1] · **val_loss** 5.465096 [1] · **val_room_acc** 0.04802693 [1] · **val_room_miou** 0.01348771 [1] · **val_struct_miou** 0.01180519 [1] · **val_wall_ext_iou** 0.03166606 [1] · **val_wall_int_iou** 0 [1] · **val_win_iou** 0 [1]

Initial 15-epoch baseline · FINISHED

50ab5d70d4b14f108a0fbdd4b95e77ca · feature/floorplan-segmentation-3d · f59bd2f · 15 epochs configured

learning_rate 1e-06 [15] · **test_door_iou** 0.103307 [0] · **test_room_acc** 0.7074836 [0] · **test_room_miou** 0.3493289 [0] · **test_struct_miou** 0.165889 [0] · **test_val_loss** 2.500442 [0] · **test_wall_ext_iou** 0.2342946 [0] · **test_wall_int_iou** 0.1022143 [0] · **test_win_iou** 0.2237402 [0] · **train_loss** 2.417938 [15] · **train_room_loss** 0.9966113 [15] · **train_struct_loss** 1.620649 [15] · **val_door_iou** 0.1021731 [15] · **val_loss** 2.442957 [15] · **val_room_acc** 0.7297459 [15] · **val_room_miou** 0.369535 [15] · **val_struct_miou** 0.1696865 [15] · **val_wall_ext_iou** 0.2236782 [15] · **val_wall_int_iou** 0.100592 [15] · **val_win_iou** 0.2523028 [15]

Earlier 25-epoch experiment · FINISHED

39254ff53b8e4ee8b153dc4067599a99 · feature/floorplan-segmentation-3d · 186c8fb · 25 epochs configured

learning_rate 1e-06 [25] · **test_door_iou** 0.004014774 [0] · **test_merged_wall_iou** 0.1207703 [0] · **test_room_acc** 0.1001274 [0] · **test_room_miou** 0.007758986 [0] · **test_struct_miou** 0.02247029 [0] · **test_val_loss** 5.421944 [0] · **test_wall_ext_iou** 0.08349573 [0] · **test_wall_int_iou** 0.001763002 [0] · **test_win_iou** 0.0006076555 [0] · **train_loss** 4.420513 [25] · **train_room_loss** 2.527551 [25] · **train_struct_loss** 2.398472 [25] · **val_door_iou** 0.005411091 [25] · **val_loss** 4.807528 [25] · **val_merged_wall_iou** 0.07485442 [25] · **val_room_acc** 0.1737167 [25] · **val_room_miou** 0.008978379 [25] · **val_struct_miou** 0.01619535 [25] · **val_wall_ext_iou** 0.05726947 [25] · **val_wall_int_iou** 0.001686902 [25] · **val_win_iou** 0.0004139263 [25]

Extended 60-epoch attempt · RUNNING

3a3605b68f3244d1af8edc4cb6f3eda5 · feature/floorplan-segmentation-3d · 186c8fb · 60 epochs configured

learning_rate 0.0002981594 [3] · **train_loss** 2.914618 [3] · **train_room_loss** 1.407941 [3] · **train_struct_loss** 1.788265 [3] · **val_door_iou** 0.08425763 [3] · **val_loss** 2.796498 [3] · **val_merged_wall_iou** 0.1841518 [3] · **val_room_acc** 0.6426879 [3] · **val_room_miou** 0.2660042 [3] · **val_struct_miou** 0.146116 [3] · **val_wall_ext_iou** 0.2053721 [3] · **val_wall_int_iou** 0.0862338 [3] · **val_win_iou** 0.2086005 [3]

12 / MLFLOW

Floorplan MLflow · alignment

floorplan-multitask-unetpp

Latest recorded values in original logged units. Brackets give the logged step. Step 0 is common for one-off evaluation imports. Run IDs and source tags are shown under each heading.

Completed 60-epoch baseline · FINISHED

6bec9eef1968482098167110c9fbc58c · feature/floorplan-segmentation-3d · 5420e12 · 60 epochs configured

learning_rate 2.829263e-06 [60] · **test_door_iou** 0.1075401 [0] · **test_merged_wall_iou** 0.223966 [0] · **test_room_acc** 0.7129684 [0] · **test_room_miou** 0.3790316 [0] · **test_struct_miou** 0.1785095 [0] · **test_val_loss** 2.46138 [0] · **test_wall_ext_iou** 0.2433502 [0] · **test_wall_int_iou** 0.1296233 [0] · **test_win_iou** 0.2335243 [0] · **train_loss** 1.801686 [60] · **train_room_loss** 0.5492282 [60] · **train_struct_loss** 1.362304 [60] · **val_door_iou** 0.108097 [60] · **val_loss** 2.573358 [60] · **val_merged_wall_iou** 0.2100824 [60] · **val_room_acc** 0.7178485 [60] · **val_room_miou** 0.4097362 [60] · **val_struct_miou** 0.1779005 [60] · **val_wall_ext_iou** 0.2255596 [60] · **val_wall_int_iou** 0.1279354 [60] · **val_win_iou** 0.2500101 [60]

Aligned 25-epoch attempt · FAILED

0dfc70e5ca2a4bb0b5b7e4ccd8171389 · feature/floorplan-segmentation-3d · 7f8cc22 · 25 epochs configured

No metrics recorded.

Aligned one-epoch smoke run · FINISHED

4ee5f64d08ad4ec49b0731f099897a49 · feature/floorplan-segmentation-3d · 7f8cc22 · 1 epochs configured

learning_rate 1e-06 [1] · **test_door_iou** 0.005159535 [0] · **test_merged_wall_iou** 0.1098828 [0] · **test_room_acc** 0.07452845 [0] · **test_room_miou** 0.009373033 [0] · **test_struct_miou** 0.0317517 [0] · **test_val_loss** 5.466134 [0] · **test_wall_ext_iou** 0.0857121 [0] · **test_wall_int_iou** 0.02370998 [0] · **test_win_iou** 0.01242517 [0] · **train_loss** 5.606272 [1] · **train_room_loss** 3.058338 [1] · **train_struct_loss** 3.159601 [1] · **val_door_iou** 0.004090879 [1] · **val_loss** 5.513886 [1] · **val_merged_wall_iou** 0.08651587 [1] · **val_room_acc** 0.1229497 [1] · **val_room_miou** 0.01074086 [1] · **val_struct_miou** 0.02465032 [1] · **val_wall_ext_iou** 0.06349896 [1] · **val_wall_int_iou** 0.02382929 [1] · **val_win_iou** 0.007182155 [1]

Aligned 25-epoch run · FINISHED

112b8d1059bb449398b3087edbf86347 · feature/floorplan-segmentation-3d · 7f8cc22 · 25 epochs configured

learning_rate 1e-06 [25] · **test_door_iou** 0.6238115 [0] · **test_merged_wall_iou** 0.7431945 [0] · **test_room_acc** 0.8583995 [0] · **test_room_miou** 0.5472586 [0] · **test_struct_miou** 0.6797542 [0] · **test_val_loss** 0.9828976 [0] · **test_wall_ext_iou** 0.738906 [0] · **test_wall_int_iou** 0.6234523 [0] · **test_win_iou** 0.732847 [0] · **train_loss** 0.7894354 [25] · **train_room_loss** 0.3575834 [25] · **train_struct_loss** 0.5033687 [25] · **val_door_iou** 0.6137536 [25] · **val_loss** 0.9666576 [25] · **val_merged_wall_iou** 0.7352128 [25] · **val_room_acc** 0.8661269 [25] · **val_room_miou** 0.6007001 [25] · **val_struct_miou** 0.6762376 [25] · **val_wall_ext_iou** 0.730659 [25] · **val_wall_int_iou** 0.6170029 [25] · **val_win_iou** 0.7435349 [25]

Later refinement attempt · RUNNING

cc26ef5775d4408797bc26240c99d33c · experiment/model-refinement-and-robustness · 4931cb9 · 15 epochs configured

learning_rate 9.891831e-05 [13] · **train_loss** 1.886589 [13] · **train_room_loss** 1.277751 [13] · **train_struct_loss** 0.8643882 [13] · **val_door_iou** 0.5591155 [13] · **val_loss** 1.69441 [13] · **val_merged_wall_iou** 0.7156078 [13] · **val_room_acc** 0.7778498 [13] · **val_room_miou** 0.4200611 [13] · **val_struct_miou** 0.6094861 [13] · **val_wall_ext_iou** 0.6184192 [13] · **val_wall_int_iou** 0.5731705 [13] · **val_win_iou** 0.6872391 [13]

13 / MLFLOW

Floorplan MLflow · refinement

CAD, Lovász, champion and fallback records

Latest recorded values in original logged units. Brackets give the logged step. Step 0 is common for one-off evaluation imports. Run IDs and source tags are shown under each heading.

CAD augmentation fine-tuning · FINISHED

ebe8add5a16341a2a258682316b0809f · feature/floorplan-segmentation-3d · 98fee5c · 12 epochs configured

composite_score 0.686531 [12] · **lr** 1e-06 [12] · **train_loss** 0.8624091 [12] · **val_door** 0.6112504 [12] · **val_loss** 0.8931896 [12] · **val_room_acc** 0.8965687 [12] · **val_room_miou_10** 0.6239396 [12] · **val_wall_ext** 0.7358352 [12] · **val_wall_int** 0.6224966 [12] · **val_wall_merged** 0.7389609 [12] · **val_win** 0.7471815 [12]

Class-weighted Lovász fine-tuning · FINISHED

d9ce1a72d3e546aaa8ad92bdb8d70d6d · feature/floorplan-segmentation-3d · de71f7d · 6 epochs configured

composite_score 67.16355 [6] · **train_loss** 0.8226559 [6] · **val_door_iou** 59.74434 [6] · **val_garage_iou** 65.54098 [6] · **val_office_iou** 0 [6] · **val_room_acc** 89.08771 [6] · **val_room_miou_10** 61.89504 [6] · **val_storage_iou** 45.73683 [6] · **val_wall_iou** 72.88338 [6] · **val_win_iou** 73.52969 [6]

Official champion benchmark import · FINISHED

a7e87552b0474bd798071baba39cc987 · release/v1.0-defense-models

latency_gpu_ms 56 [0] · **test_door_iou** 61.7767 [0] · **test_door_iou_tta** 63.21013 [0] · **test_merged_wall_iou** 74.17111 [0] · **test_merged_wall_iou_tta** 74.93261 [0] · **test_room_acc** 87.49611 [0] · **test_room_acc_tta** 88.08799 [0] · **test_room_miou_10c** 56.39748 [0] · **test_room_miou_10c_tta** 58.24419 [0] · **test_room_miou_9c** 62.66386 [0] · **test_struct_miou** 67.83129 [0] · **test_struct_miou_tta** 68.92291 [0] · **test_win_iou** 73.51749 [0] · **test_win_iou_tta** 74.65292 [0]

Fallback blend metadata import · FINISHED

cfee34c931704f16b051d5b60f7c9d93 · experiment/floorplan-style-adaptation

blend_alpha 0.3 [0] · **parity_delta_max** 6.31e-06 [0]

The champion import stores its IoUs as percentages and also logs a 56 ms GPU latency field. The benchmark JSON reports separate measured mean inference latencies. The fallback import records blend alpha and a parity delta.

14 / MLFLOW

Facade MLflow · baselines

pix2pix-cmp-facades

Latest recorded values in original logged units. Brackets give the logged step. Step 0 is common for one-off evaluation imports. Run IDs and source tags are shown under each heading.

Pix2Pix smoke attempt · FAILED

762a9ecda6b54e2c9525f601567d7630 · main · f8ac87b · 1 epochs configured

loss_D 0.7145748 [1] · **loss_G** 37.70189 [1] · **loss_pixel_train** 0.3688284 [1] · **loss_pixel_val** 0.3427441 [1]

Pix2Pix smoke attempt · FAILED

e747fe922cde484c9c4ce7b48f58a15d · main · f8ac87b · 1 epochs configured

loss_D 0.6847683 [1] · **loss_G** 37.8868 [1] · **loss_pixel_train** 0.3707974 [1] · **loss_pixel_val** 0.3787848 [1]

Pix2Pix one-epoch smoke run · FINISHED

c8ae12969b194a19a8ddcd0817b30775 · main · f8ac87b · 1 epochs configured

loss_D 0.6847683 [1] · **loss_G** 37.8868 [1] · **loss_pixel_train** 0.3707974 [1] · **loss_pixel_val** 0.3787848 [1]

Pix2Pix 100-epoch baseline · FINISHED

e463b47e8cec464ba45bf1c0e16d2d2a · main · 616e2e5 · 100 epochs configured

loss_D 0.009707542 [100] · **loss_G** 13.86999 [100] · **loss_pixel_train** 0.1288424 [100] · **loss_pixel_val** 0.3549642 [100]

Enhanced 200-epoch configuration · RUNNING

8de9e858c18442919a9f2f68bfd56482 · experiment/enhanced-pix2pix-200ep · 352a76e · 200 epochs configured

loss_D 0.002864421 [171] · **loss_G** 19.77762 [171] · **loss_pixel_train** 0.1877772 [171] · **val_l1** 0.3364877 [171] · **val_psnr** 13.14294 [171] · **val_ssim** 0.3358944 [171]

15 / MLFLOW

Facade MLflow · perceptual and 512

pix2pix-cmp-facades

Latest recorded values in original logged units. Brackets give the logged step. Step 0 is common for one-off evaluation imports. Run IDs and source tags are shown under each heading.

Perceptual 100-epoch training · FINISHED

1ea21de0f4da4738be910893bba7b2bd · experiment/perceptual-sharpness-pix2pix · 36ec4e3 · 100 epochs configured

loss_D 0.08940339 [100] · **loss_G** 12.33324 [100] · **loss_pixel_train** 0.2256806 [100] · **val_l1** 0.3504192 [100] · **val_lpips** 0.3259458 [100] · **val_psnr** 12.71814 [100] · **val_ssim** 0.2787956 [100]

Perceptual EMA 100-epoch training · FINISHED

702efec39c104b1c9848d0a71b05295d · experiment/model-refinement-and-robustness · e218e26 · 100 epochs configured

loss_D 0.04277952 [100] · **loss_G** 11.9992 [100] · **loss_pixel_train** 0.2152927 [100] · **lr_g** 3.921569e-06 [100] · **val_l1** 0.3528406 [100] · **val_lpips** 0.3251982 [100] · **val_lpips_ema** 0.3216413 [100] · **val_psnr** 12.64786 [100] · **val_ssim** 0.2695908 [100]

Promoted EMA benchmark import · FINISHED

1835cac5e55b4462ac850e14b2c418d0 · release/v1.0-defense-models

kid_x1000 63.552 [0] · **val_l1** 0.3403 [0] · **val_lpips** 0.3096 [0] · **val_psnr** 12.97 [0] · **val_ssim** 0.2893 [0]

512-pixel test and decoder history import · FINISHED

ee9e3b8af4524574a9b6efe694afd6e6 · experiment/facade-photorealism-512

device_mib 1246 [0] · **history_loss_d** 0.7711462 [5] · **history_loss_g** 14.46784 [5] · **history_loss_vgg** 1.05489 [5] · **history_val_vgg** 1.084145 [5] · **kid_x1000** 69.61554 [0] · **opening_edge_mae** 0.09493339 [0] · **peak_allocated_mib** 737.69 [0] · **val_lpips** 0.4924111 [0] · **val_psnr** 13.32883 [0] · **val_ssim** 0.3617624 [0]

Diffusion configuration import · FINISHED

ee444a71e6eb45b29671c91006508d22 · experiment/facade-diffusion-control

control_steps 30 [0] · **target_resolution_px** 1024 [0]

The EMA and 512 imports use val_ metric keys for values sourced from held-out benchmark artifacts. The 512 history metrics shown here are the latest decoder training snapshots at step 5. Its test result uses the checkpoint selected at epoch 1.

16 / REFERENCE

Metric guide and evidence

Interpretation, tracking coverage and reproducibility

Metric definitions

Metric	How to read it
IoU / mIoU ↑	Intersection divided by union of pixels. mIoU averages over a specified class set. Confirm the included classes before comparing.
Precision / recall / F1 ↑	Precision is the share of predictions matched. Recall is the share of references matched. F1 is their harmonic mean. Matching varies by protocol.
Room pixel accuracy ↑	Correct room-class pixels divided by evaluated pixels. Class imbalance can favor this metric relative to mIoU.
Shell validity ↑	Share of plans passing the specific geometry or repair checks. Use room accuracy and external usability alongside it.
MAE ↓ / signed bias	MAE is mean absolute error. Signed count bias is prediction minus ground truth. Positive and negative mistakes can cancel in bias.
L1 ↓ / opening-edge MAE ↓	Average absolute error in image values or in edge responses around openings, respectively.
SSIM ↑ / PSNR ↑	Structural similarity and peak signal-to-noise ratio. PSNR is measured in dB.
LPIPS ↓ / KID ↓	LPIPS measures paired perceptual feature distance. KID compares feature distributions. Compare within the same extractor, scale and sampling protocol.
Latency ↓ / memory ↓	Reported in milliseconds and MiB. Device-sampled and framework peak counters capture different quantities.

MLflow model versions

Registered name	Version	Training run	Stage	Status
ArchiGen-Pix2Pix-Generator	1	c8ae1296	None	READY
ArchiGen-Pix2Pix-Generator	2	e463b47e	None	READY
ArchiGen-Pix2Pix-Perceptual	1	1ea21de0	None	READY
ArchiGen-Pix2Pix-Perceptual	2	702efec3	None	READY

The snapshot records 17 FINISHED, 4 FAILED and 3 RUNNING runs. RUNNING is the stored status and does not establish active execution. The Default experiment has no runs. An imported score and its originating evaluation count as one measurement.

Evidence inside the PDF

The attachments pane contains a ZIP with all latest metrics, historical points, recorded run parameters, relevant provenance tags, registry rows, Git refs and benchmark JSONs. The source manifest distinguishes GitHub snapshots and local results and includes the original SHA-256 of each JSON. The local database was read in read-only mode.

The audit covers the fetched public branches and the MLflow SQLite store and saved evaluations available in this workspace. No external tracking server or missing historical artifacts were assumed. No training or benchmark was rerun for this report.

Evidence · model_registry.json · README.md · defense_runbook.md

17 / SUPPLEMENT

Additional saved diagnostics

Historical figures, external blend ablations and geometry smoke checks

Historical floorplan figures published in README

Metric group	Reported result
Structure IoUs	Merged walls 74.32%, exterior walls 73.89%, interior walls 62.34%, doors 62.38%, windows 73.28%.
Room scores	85.85% pixel accuracy and 54.72% semantic mIoU.
Shell validity	96.25% watertight shells. The official test and routing tables in this report use their own protocols.
Exterior wall classification	91.07% accuracy, 94.21% precision and 92.41% recall. The runbook describes a test-model check. Full per-wall labels were unavailable.

The README calls the radius-based door results test-split results. The saved script loads the validation split, so this report labels the figures as validation. The README separately reports a 72% to 75% per-image win rate for the paired facade study with $p < 10^{-6}$. The underlying paired statistics were unavailable for verification.

Development blend sweep · 20 plans

Variant	Valid shells	Zero-room plans	Detected rooms	Mean rooms
Champion v1.1	20	0	118	5.90
Blend a=0.3	20	0	127	6.35
Blend a=0.5	19	0	158	7.90
Blend a=0.7	20	0	166	8.30
Challenger	20	0	188	9.40

This sweep recorded 127 rooms and 20 valid shells for α=0.3. The separate development triage run recorded 111 rooms and 19 shells for that alpha. The attachment preserves both snapshots and their per-plan records.

Targeted geometry and facade checks

Check	Recorded result
Outdoor class in footprint	399 valid plans. With vs without Outdoor, footprint IoU 78.59% vs 75.54% and mean perimeter error 21.29% vs 23.89%.
Targeted fallback checks	5 real plans. Champion 2 valid shells, fallback 4. PyTorch and ONNX each triggered the fallback 2 times.
Geometry branch browser smoke	Two levels rendered at 0 and 2.8 m on the traced sample. Separate detected fixture hosted 15 openings and left 7 unresolved.
512 validation diagnostic	First 16 images, CPU FP32, epoch 1. VGG loss was 1.10539 for the prior and 1.07585 for the experiment.

Evidence · README.md · dev_blends_comparison.json (local evidence) · perimeter_outdoor_hypothesis_analysis.json (local evidence) · fallback_real_plans_eval.json (local evidence) · browser-verification.json (local evidence) · validation_diagnostic.json (local evidence)

18 / ARCHITECTURE

Floorplan model architecture

ResNet-34 UNet++ with shared features and two segmentation heads

Plan input
512 × 512 RGB
Aspect-preserving padding

→

Shared encoder
ResNet-34
ImageNet initialization

→

Shared decoder
UNet++ nested skips
Dense spatial features

→

Two pixel heads
5 structural logits
12 room logits

What the network predicts

Component	Implementation
Structure head	Five classes. Background, interior wall, exterior wall, door and window.
Room head	A separate 3 × 3 segmentation head on the shared decoder output. Twelve classes cover background, LivingRoom, Bedroom, Kitchen, Bath, Entry, Outdoor, Storage, Closet, Garage, Office and Other.
Supervision	CubiCasa5K RGB plans with SVG-derived structure and room masks. The aligned loader applies the same scale and pad offsets to image and labels. Standard training uses class-weighted cross-entropy and foreground Dice in each head.
Loss balancing	The standard trainer weights structure Dice by 1.5 and the whole room objective by 0.8. Structural class weighting gives doors more weight than either wall class.

How the masks become a 3D house

Class masks
Argmax of logits
Optional TTA first

→

Vectorization
Walls, openings, rooms
Envelope and scale

→

Building assembly
Floor and wall meshes
Roof and level placement

The vectorizer and viewer apply geometric rules after segmentation. They convert the plan’s horizontal coordinates into Three.js X and Z and use Y for height. Gaps, room fragments or missed openings can propagate from the predicted masks. Geometry checks and a fallback can recover some failures.

The default checkpoint uses the same dual-head network on PyTorch or ONNX. Four-fold TTA averages inverse-transformed probabilities. The alternate checkpoint is a weight blend in the same architecture. The router tries it for zero rooms, an invalid shell, or a lone room under 8 m^2, then applies acceptance checks to the alternate layout.

Evidence · models.py · dataset.py · train.py · floorplan_engine.py · houseGeometry.js

19 / ARCHITECTURE

Facade model architecture

Registered 256-pixel Pix2Pix EMA generator and the wall-to-texture bridge



256-pixel reference model

Component	Implementation
Encoder and bottleneck	Eight stride-2 convolution blocks shrink 256 × 256 to a 1 × 1 bottleneck. The 64-channel first stage grows to 512 channels. Intermediate stages use instance normalization and leaky ReLU.
Decoder and output	Seven skip-connected upsampling blocks and a final RGB projection. The promoted model uses nearest-neighbor upsampling plus convolution, with Tanh output in [−1, 1].
Training discriminator	One conditional 70 × 70 PatchGAN judges the semantic sketch concatenated with either the real or generated RGB photo. Its intermediate activations are also used for feature matching.
Promoted training recipe	Least-squares GAN, L1 ×30, SqueezeNet LPIPS ×10 and discriminator feature matching ×10. EMA decay 0.999 and linear learning-rate decay from epoch 50. Inference loads the EMA generator alone.

Conditioning and style

The bridge encodes wall dimensions, opening positions and trim into the CMP semantic palette. Aspect-preserving padding and returned UV bounds connect the texture to the physical wall. The Three.js viewer uses planar UVs and positions frames from the same facade geometry.

The 256-pixel network is conditioned on the sketch. The named material choices in the historical bridge apply image post-processing. Architectural style classes were not supervised in CMP.

What the benchmark evaluates

The historical CMP matrix measures generated images against paired facade photos. Viewer checks measure alignment of semantic openings and frames. Both are useful for diagnosing the pipeline, and they measure different outcomes.

Evidence · architecture.py · train.py · losses.py · facade_bridge.py · facadeGeometry.js

20 / ARCHITECTURE

512 refiner and diffusion pilot

Two experimental paths that preserve the application’s wall geometry



512-pixel residual FiLM GAN

The refiner has a 24-channel stem, three downsampling stages, four bottleneck residual blocks and three skip-connected decoder stages. A 32-dimensional style embedding predicts channel-wise scale and shift after instance normalization. The initial residual head starts at zero so the model starts from the coarse prior.

Two independent PatchGANs see paired images at 512 and downsampled 256. The final recipe uses adversarial loss ×1, L1 ×5, VGG-19 perceptual ×10, feature matching ×5 and opening-edge loss ×5. VGG uses relu1_2 through relu5_2. The edge term compares Sobel and Laplacian responses near the semantic window and door masks.

The selected trial fine-tuned late coarse decoder blocks at 0.1× the main learning rate. FP16 autocast, gradient checkpointing and four-step accumulation gave an effective batch of eight from physical batches of two. The reported device peak across training trials was 2,342 MiB. CMP supported only the neutral embedding.

1024-pixel diffusion and geometry-locked composite



Whole-facade diffusion trials shifted or invented openings. The retained prototype uses a reviewed SDXL material sample and deterministic wall tiling and opening masks. It uses a pretrained model with CPU or disk offload, and no new style training or held-out quality benchmark was recorded.

Evidence · architecture.py · losses.py · generator.json · facade512-experiment.md · facade-diffusion-experiment.md

21 / EXPERIMENT CHANGES

Floorplan training experiments

What changed in the data, objective or training budget and what each run measured

Experiment and reference	What changed	Recorded outcome
Early baseline and smoke runs 0b67f07 · 0e816836, 832abe91, 50ab5d70	Introduced the shared ResNet-34 UNet++ with 5 structure and 12 room outputs, training at 512 pixels. One-epoch runs checked the training path before the 15-epoch baseline.	First smoke attempt failed and the second finished. The 15-epoch test reached 10.33% door IoU and 34.93% room mIoU.
Longer pre-alignment training 186c8fb · 39254ff5, 3a3605b6, 6bec9eef	Tried 25 and 60 epochs with the same architecture before the label alignment fix. The branch also added a 400-plan evaluator and an early hybrid fallback.	The first 25-epoch run gave 12.08% merged-wall IoU and 0.40% door IoU. The completed 60-epoch run reached 22.40% and 10.75%. The other 60-epoch attempt stopped logging at step 3.
Aligned 25 epochs b0b8399 · 0dfc70e5, 4ee5f64d, 112b8d10	Applied the image letterbox transform to SVG target coordinates and added split-specific NPZ caching. Trained the same network with physical batch 2, then exported ONNX.	One failed attempt and one smoke run preceded the completed model. Test wall IoU 74.32%, door IoU 62.38%, window IoU 73.28%, room mIoU 54.73%.
CAD fine-tune finetune.py · ebe8add5	Resumed the aligned checkpoint for 12 epochs. Used seven CAD appearance augmentations at 50% total probability, froze encoder batch-normalization layers, and used cosine learning-rate decay.	Latest validation 10-class room mIoU 62.39%. This checkpoint became the registered floorplan champion.
Lovász and class weights finetune_lovasz.py · d9ce1a72	Fine-tuned the CAD checkpoint for six epochs. Added Lovász-Softmax to cross-entropy and Dice in both heads, with capped inverse-frequency room weights.	Latest validation wall IoU 72.88%, door IoU 59.74%, room mIoU 61.90%. Office IoU stayed at 0%.
Higher input resolution 883ee19 · cc26ef57	The refinement branch logged a 1024-pixel UNet++ training configuration with 15 epochs planned and learning rate 0.0001.	Last snapshot at step 13. Validation wall IoU 71.56%, door IoU 55.91%, room mIoU 42.01%. No completed test result was recorded.

The 60-epoch and aligned 25-epoch runs differ in training setup as well as label alignment. Their scores show the size of the recorded gain across those runs. The experiments do not isolate the effect of each change.

Evidence · train.py · Alignment patch · Checkpoint registry

22 / EXPERIMENT CHANGES

Floorplan robustness experiments

Input style adaptation, test-time inference and downstream geometry

Experiment and reference	What changed	Recorded outcome
Procedural CAD restyle 216387f · acfd336	Rendered training SVGs with varied line weights, hatching, blueprints and dimensions while preserving masks. Fine-tuned the same champion backbone. The saved run contains one epoch.	On 12 locked external plans, valid shells rose from 9 to 12 and room-count MAE fell from 5.25 to 4.42. Official test room mIoU fell from 56.40% to 52.30%.
Blend and routing f23e02e · cfee34c9	Interpolated champion and restyled model weights at α=0.3, 0.5 and 0.7. Registered the α=0.3 blend as a fallback for collapsed layouts and exposed API telemetry.	The routed system used fallback for 5 of 400 official test plans. Valid shells rose from 99.25% to 99.75%. Room count MAE moved from 1.562 to 1.575.
Test-time augmentation de71f7d · f068c95 · a7e87552	Averaged inverse-transformed probability maps from original and flipped inputs. A D4 eight-transform option was also implemented. No retraining was needed.	Four-fold TTA raised official test room mIoU from 56.40% to 58.24%. Mean inference time rose from 104 to 328 ms.
Topology and export e218e26 · 11499e8	Added polygon validity repair and exported the champion and blend to ONNX. Verified the exported and PyTorch layouts over the same plans.	Mean room overlap was 98.39% and label agreement was 100% over 50 parity plans. Validation shell rates varied with the repair protocol.
House geometry reconstruction d608569 · 7bcf84a	Local experiment ref for joined wall assemblies, opening hosting, clipped floors, aligned levels and separate roof geometry. Changes happen after segmentation.	The saved smoke check placed two traced levels at 0 and 2.8 m. A separate detected fixture had 15 hosted and 7 unresolved openings. No new segmentation training run was identified.

Evaluation protocol changes

The refinement work also locked a 12-plan external set by SHA-256 and added blind evaluation. The source tagged a7e87552 is an official champion benchmark import. The fallback record cfee34c9 logs blend and parity metadata. Neither adds a training run.

Evidence · train_restyled.py · floorplan_engine.py · evaluate_test_authoritative.py · Locked evaluation suite

23 / EXPERIMENT CHANGES

Facade experiment progression

Architecture, training and evaluation changes from the baseline to the diffusion pilot

Experiment and reference	What changed	Recorded outcome
Baseline Pix2Pix 100 epochs 77313d5 · e463b47e	Trained a 256-pixel conditional U-Net with transposed-convolution decoder blocks and a single PatchGAN. Adversarial loss plus L1 ×100. One-epoch smoke runs preceded the full run.	Historical CMP test LPIPS 0.6327 and KID ×1000 150.357. The baseline MLflow training run logged losses.
Enhanced Pix2Pix 352a76e · 8de9e858	Replaced transposed convolutions in decoder blocks with nearest-neighbor resize and convolution. Added paired scale jitter, crops and flips, plus validation SSIM and PSNR. Retained L1 ×100.	Retained 150-epoch checkpoint reached LPIPS 0.4940, SSIM 0.3345 and KID ×1000 109.857. The 200-epoch configuration last logged at step 171.
Perceptual sharpness 36ec4e3 · 1ea21de0	Kept the resize-convolution generator. Lowered L1 to 30, added SqueezeNet LPIPS ×10 and PatchGAN feature matching ×10, and disabled color jitter.	Historical LPIPS improved to 0.3169 and KID ×1000 to 67.649. SSIM fell to 0.2792, reflecting a trade-off in the measured objectives.
EMA and schedule 6620891 · 702efec3, 1835cac5	Kept the perceptual architecture and losses. Added an exponential moving average of generator weights with decay 0.999 and linear learning-rate decay from epoch 50.	Promoted 256-pixel checkpoint reached LPIPS 0.3096 and KID ×1000 63.552 in the historical comparison.
512 residual FiLM GAN 3dc23fc · ee9e3b8a	Added the native 512 sketch refiner and two discriminator scales. Compared a frozen coarse prior, stronger VGG and edge weights, then a low-rate fine-tune of late coarse decoder blocks.	Selected decoder epoch 1. Paired 512-protocol LPIPS fell from 0.49887 to 0.49241 and edge MAE from 0.10771 to 0.09493. Neutral CMP only.
SDXL diffusion control 791d2b3 · ee444a71	Tested 1024-pixel Canny and prompt controls with memory offload. Rejected geometry-drifting outputs and used a reviewed brick swatch in a fixed-opening material composite.	The retained prototype kept five prescribed openings. The result is a qualitative and viewer check. No held-out CMP score was recorded.

The 512 experiment was evaluated with 512 targets and its own KID extractor. Compare it to the paired EMA row in the 512 benchmark. The 256 values above come from the historical comparison.

Evidence · facade_authoritative_metrics.json · facade512-experiment.md · facade-diffusion-results.json

24 / EXPERIMENT CHANGES

Ablations and scope

Individual 512 trials and related branches without new two-model training runs

512 ablation	What changed	Best validation VGG
Initial residual head	Froze the 256 prior. Optimized the residual head with VGG weight 1 and opening-edge weight 2. Four epochs recorded.	1.12398 at epoch 1
Higher perceptual and edge weights	Warm-started from the initial checkpoint. Increased VGG from 1 to 10 and edge weight from 2 to 5. Five epochs recorded.	1.08337 at epoch 1
Coarse decoder tuning	Started from the stronger-loss checkpoint. Trained late prior decoder blocks at one-tenth the main learning rate. Five epochs recorded.	1.05881 at epoch 1

The recorded 512 trials were sequential warm starts. The validation VGG series guides checkpoint selection. A single-variable causal effect cannot be assigned from these runs. Only the selected decoder checkpoint was evaluated on the held-out test.

Other published experiment branches

Branch or workstream	What changed	Evaluation scope
studio-interface-design 17b9686	Refined the responsive Floorplan Studio and Generative Studio interfaces.	UI change. No new floorplan or facade model training was identified.
furniture-reconstruction-quality 5b32a0a	Added an isolated image-to-3D furniture quality path, texture baking and mesh checks.	Separate furniture model family. Its measurements are outside this two-model report.
Champion benchmark imports a7e87552 · 1835cac5	Recorded existing floorplan and facade benchmark scores in MLflow for the promoted checkpoints.	Previously measured results, counted once.
Training smoke runs See the MLflow register	Exercised one-epoch training paths for both families and checked aligned floorplan loading.	Pass, failure and partial histories are retained. These runs provide no final comparative checkpoint result.

How to use the original report

The original evaluation matrices, run register and embedded ZIP are preserved. Use the PDF bookmarks to jump between these architecture notes and the benchmark pages. Branch links identify code changes. MLflow run IDs identify the recorded execution.

Evidence · experiment_results.json · furniture-quality-experiment.md · model_registry.json

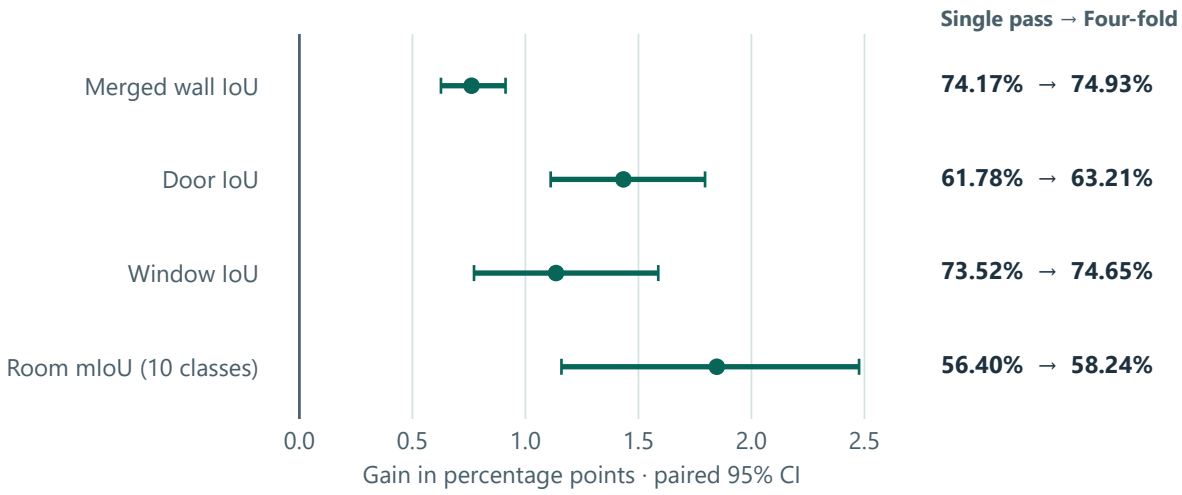
25 / EVALUATION CHARTS

Floorplan accuracy and training

Test-time augmentation and the SVG alignment correction

Four-fold TTA on the official test set

Paired differences on 400 CubiCasa5K plans. Right of zero favors TTA.



MEAN INFERENCE PER PLAN

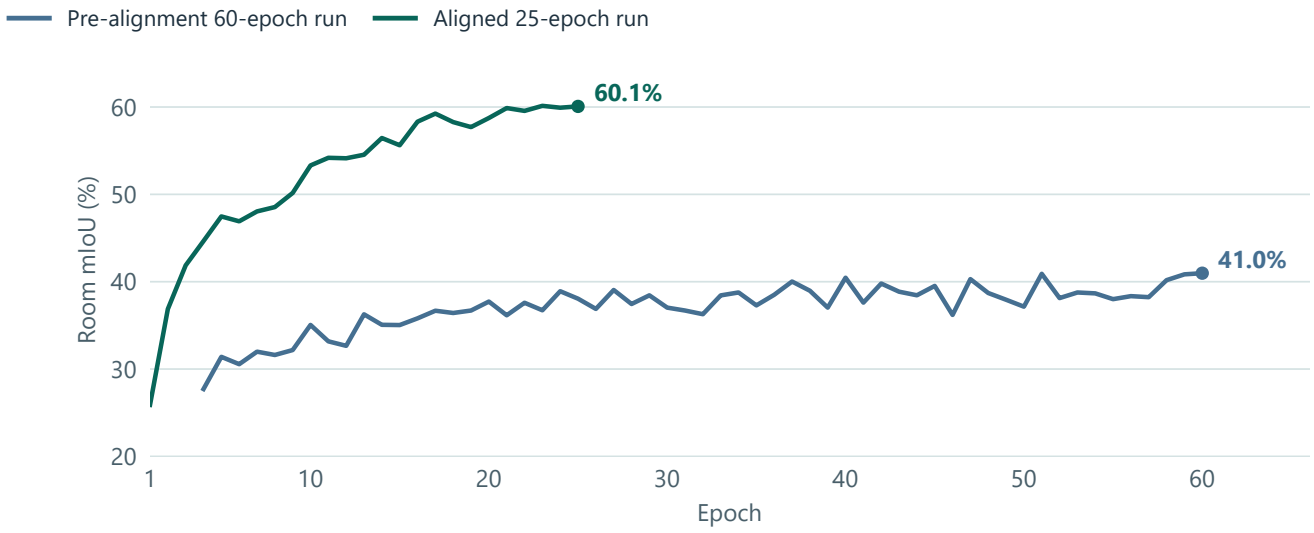
104 ms Single pass

MEAN INFERENCE PER PLAN

328 ms Four-fold TTA

Validation room mIoU during training

Two saved MLflow runs with different training setups.



The pre-alignment trace starts at the first available logged epoch (4).

Data · Official test benchmark on page 2 · MLflow runs 6bec9eef and 112b8d10

26 / EVALUATION CHARTS

Floorplan generalization

Shell validity, count errors and usability on external plans

Locked external set

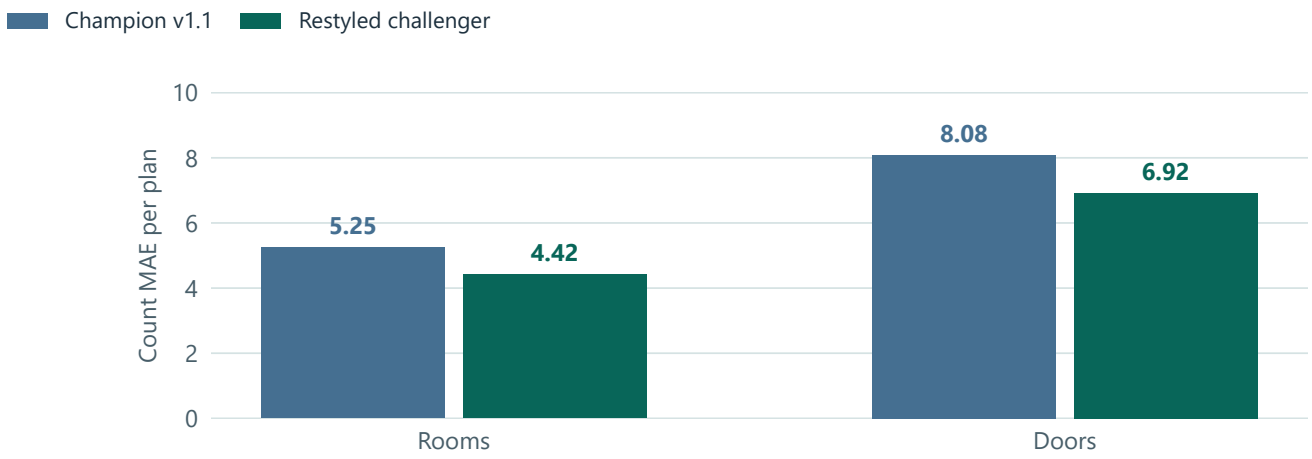
12 plans · Champion v1.1 compared with the restyled challenger

CHAMPION V1.1

9/12 Valid shells

RESTYLED CHALLENGER

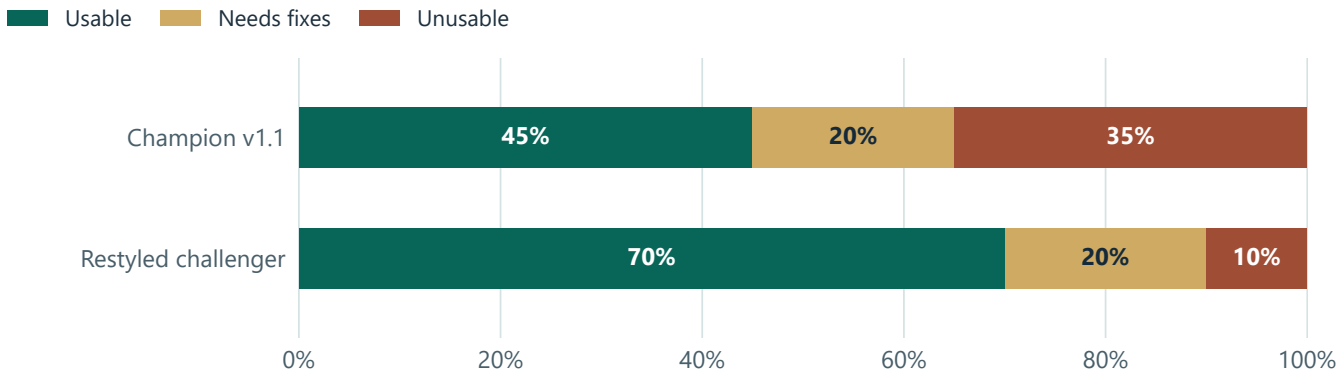
12/12 Valid shells



Lower mean absolute error (MAE) means the estimated counts were closer to the labels.

Blind usability ratings

Separate 20-plan development set · each bar represents all 20 plans



The locked external set and the development ratings are separate evaluations.

Data · External generalization on page 6 · Locked plan reports and blind ratings in the attachment

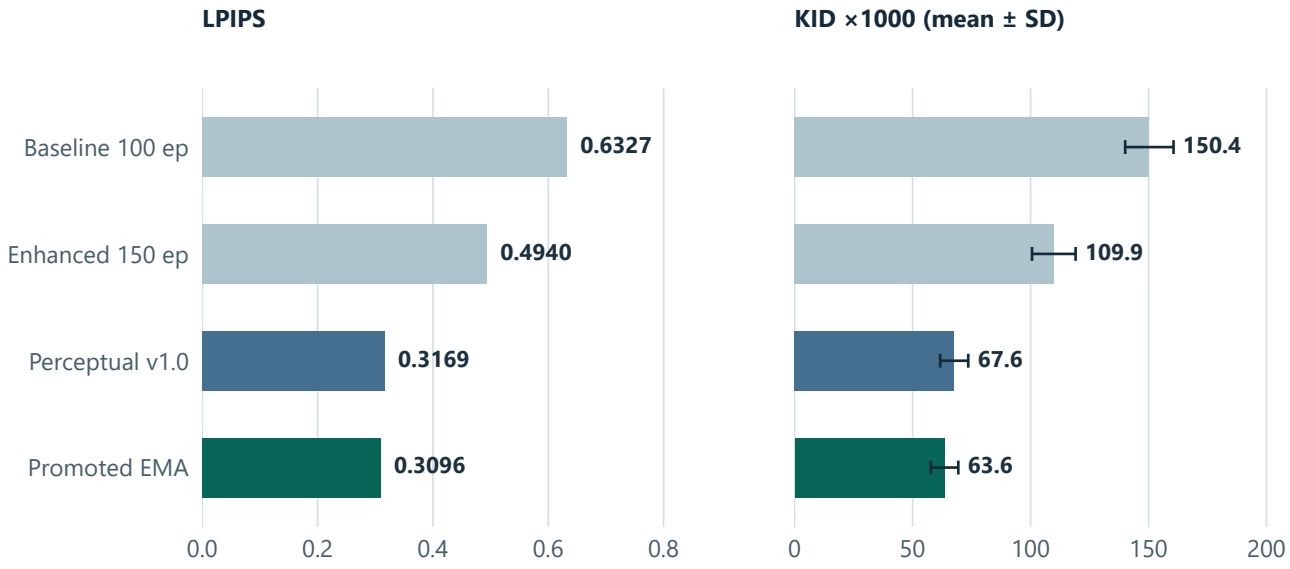
27 / EVALUATION CHARTS

Facade checkpoint comparison

Historical 256-pixel CMP test and the EMA training trace

Four retained checkpoints

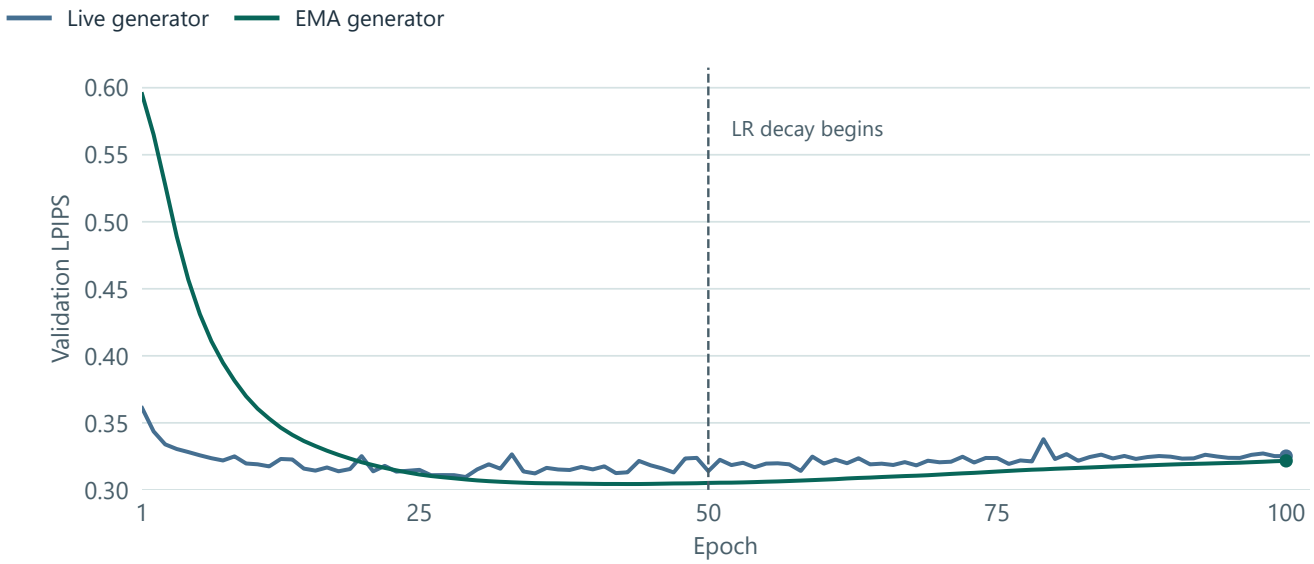
Historical CMP test · 106 images · lower is better for both metrics



KID whiskers show standard deviation across the 100 subset draws.

Validation LPIPS within the EMA training run

Live generator weights and their exponential moving average, logged on the same epochs.



Validation traces and held-out checkpoint scores come from different splits.

Data · 256-pixel benchmark on page 8 · MLflow training run 702efec3

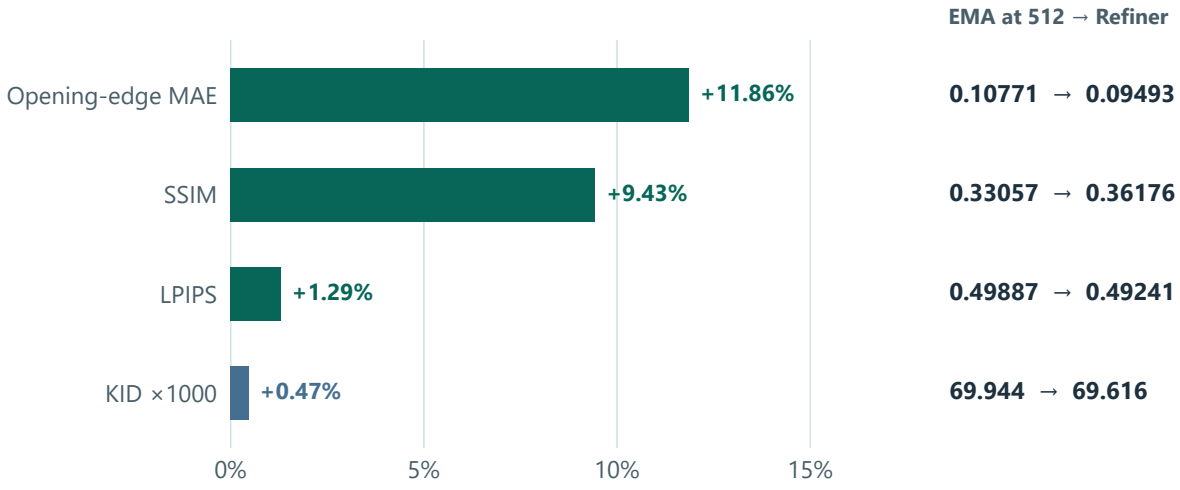
28 / EVALUATION CHARTS

Facade 512-pixel results

Paired CMP quality metrics and the refiner's validation trials

Quality change against the upsampled EMA baseline

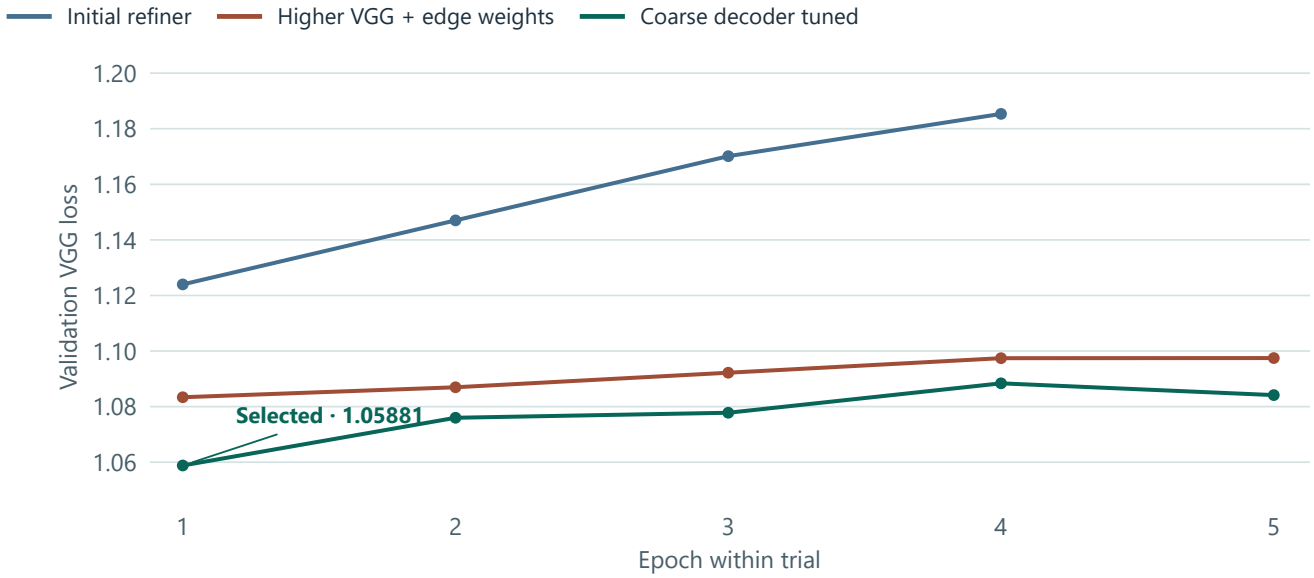
106 paired test images at 512 pixels · positive bars mean improvement



LPIPS difference, refiner minus EMA · paired 95% CI [-0.01014, -0.00264].
PSNR 13.069 → 13.329 dB. The small KID change is descriptive.

Validation VGG loss across three 512 training trials

Sequential warm starts · the decoder's first epoch was selected before test evaluation



The loss curves guide checkpoint selection within this experiment.

Data · Paired CMP benchmark on page 9 · Saved refiner training histories on page 10